Ordinary Differential Equations Note

Differential Equations with Boundary-Value Problems, 9/e (Metric Edition)

2 First-Order Differential Equations

2.1 Solution Curves without a Solution

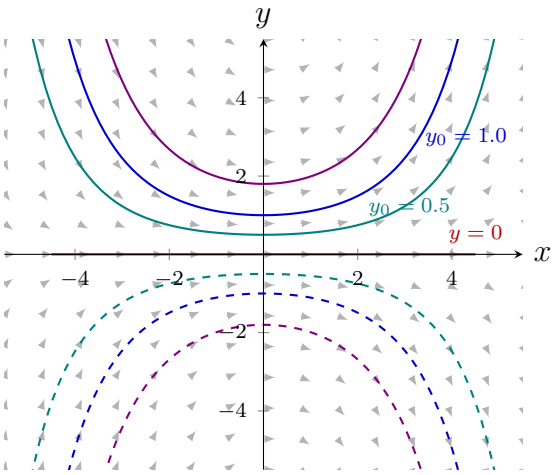
2.1.1 Direction Field(方向場)

Find solutions with the direction field is a qualitative analysis(定性分析).

Example. The equation

$$y' = 0.2xy.$$

With the direction field and solution curve:



2.1.2 Phase portrait for Autonomous ODE(自主性方程與相圖)

To solve the equation shown as the form

$$\frac{dy}{dx} = f(y),$$

we will use the following steps:

1. Equilibrium(平衡) / Critical Points: Let $y' = f(y) = 0$ and find its critical points.

   Example.

   $$\frac{dp}{dt} = p(a - bp) = f(p).$$

   Let $p' = p(a - bp) = f(p) = 0$, implies

   $$p(a - bp) = 0.$$

   Then $p = 0$ or $p = \frac{a}{b}$.

2. Phase portrait

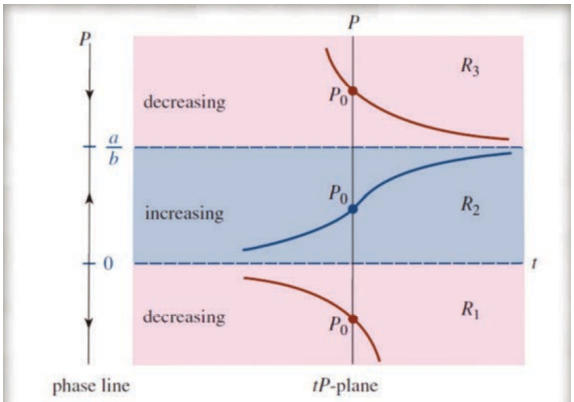


FIGURE 2.1.6 Phase portrait and solution curves in each of the three subregions in Example 4

The **attractor**(吸引子) is a point that both arrows on either side of the phase line point to this point, which is the stable equilibrium point (e.g. $p = \frac{a}{b}$); and the **repeller**(排斥子) is a point that the arrows on both sides of the phase line are moving away from the point, indicating an unstable equilibrium point. (e.g. $p = 0$)

2.2 Separable Equations

If $\frac{dy}{dx} = g(x)$ and $g(x)$ is continuous, then

$$\begin{aligned} \frac{dy}{dx} = g(x) &\implies \frac{dy}{dx}dx = g(x)dx \\ &\implies dy = g(x)dx \\ &\implies \int dy = \int g(x)dx + C \\ &\implies y = \int g(x)dx + C \end{aligned}$$

Definition. A first-order differential equation form

$$\frac{dy}{dx} = g(x)h(y)$$

or

$$\frac{dy}{dx} = \frac{g(x)}{h(y)}$$

or

$$\frac{dy}{dx} = \frac{h(y)}{g(x)}$$

is said to be **separable** or to have separable variables.

Case 1: If $h(y) \neq 0$[1], then

$$\begin{aligned}
\frac{dy}{dx} = g(x)h(y) &\implies \frac{1}{h(y)}dy = g(x)dx \\
&\implies \int \frac{1}{h(y)}dy = \int g(x)dx + C,\ C \in \mathbb{R} \\
&\implies \cdots \\
&\implies y = \Phi(x) + \xi
\end{aligned}$$

and

$$\begin{aligned}
\frac{dy}{dx} = \frac{g(x)}{h(y)} &\implies h(y)dy = g(x)dx \\
&\implies \int h(y)dy = \int g(x)dx + C,\ C \in \mathbb{R}
\end{aligned}$$

Case 2: If $g(x) \neq 0$ and $h(y) \neq 0$, then

$$\begin{aligned}
\frac{dy}{dx} = \frac{h(y)}{g(x)} &\implies \frac{1}{h(y)} = \frac{1}{g(x)}dx \\
&\implies \int \frac{1}{h(y)} = \int \frac{1}{g(x)}dx + C,\ C \in \mathbb{R}
\end{aligned}$$

Example. Solve the equation

$$y' = \frac{y}{1+x}.$$

Sol.

$$\begin{aligned}
y' = \frac{dy}{dx} = \frac{y}{1+x} &\implies \frac{1}{y}dy = \frac{1}{1+x}dx \\
&\implies \int \frac{1}{y}dy = \int \frac{1}{1+x}dx + C \\
&\implies \ln|y| = \ln|1+x| + C \\
&\implies e^{\ln|y|} = e^{\ln|1+x|+C} \\
&\implies y = e^C(1+x),\ C \in \mathbb{R}
\end{aligned}$$

[1]For the form $\frac{dy}{dx} = g(x)h(y)$ and $\frac{dy}{dx} = \frac{g(x)}{h(y)}$.

Example. Solve the equation

$$y' = \frac{1}{1+\sin x}.$$

Sol.

$$\begin{aligned}
y' = \frac{dy}{dx} = \frac{1}{1+\sin x} &\implies dy = \frac{1}{1+\sin x}dx \\
&\implies \int dy = \int \frac{1}{1+\sin x}dx + C \\
&\implies y = \int \frac{1}{1+\sin x}\cdot\frac{1-\sin x}{1-\sin x}dx + C \\
&\implies y = \int \frac{1-\sin x}{1-\sin^2 x}dx + C \\
&\implies y = \int \frac{1-\sin x}{\cos^2 x}dx + C \\
&\implies y = \int \frac{1}{\cos^2 x}dx - \int \frac{\sin x}{\cos^2 x}dx + C \\
&\implies y = \int \sec^2 x dx - \int \tan x \sec x dx + C \\
&\implies y = \tan x - \sec x + C,\ C \in \mathbb{R}.
\end{aligned}$$

Example. Solve the equation

$$\frac{dy}{dx} = y^2 - 4.$$

Sol.

$$\begin{aligned}
\frac{dy}{dx} = y^2 - 4 &\implies \int \frac{1}{y^2-4}dy = \int dx + C \\
&\implies \int \frac{1}{(y-2)(y+2)} = x + C \\
&\implies \int \frac{1/4}{y-2} + \frac{-1/4}{y+2}dy = x + C \\
&\implies \frac{1}{4}\left(\ln|y-2| - \ln|y+2|\right) = x + C.
\end{aligned}$$

Thus

$$\begin{aligned}
&\frac{1}{4}\left(\ln|y-2| - \ln|y+2|\right) = x + C \\
\implies &\ln\left|\frac{y-2}{y+2}\right| = 4x + 4C \\
\implies &\frac{y-2}{y+2} = e^{4x+4C} \\
\implies &y = 2\cdot\frac{e^{4x+4C}+1}{1-e^{4x+4C}},\ C \in \mathbb{R}.
\end{aligned}$$

So $y = 2\cdot\dfrac{e^{4x+4C}+1}{1-e^{4x+4C}}$ is a general solution of the equation.

Question: In the last example, is there any singular solution for the equation? $y = 2$ and $y = -2$ are solutions, but $y = 2$ can be derived from the general solution. Hence $y = -2$ is a singular solution.

Example. Solve

$$\cos x(e^{2y} - y)\frac{dy}{dx} = e^y \sin 2x$$

with $y(0) = 0$.

Sol.

$$\begin{aligned}
\cos x(e^{2y} - y)\frac{dy}{dx} = e^y \sin 2x &\implies \cos x(e^{2y} - y)\frac{dy}{dx} = 2e^y \sin x \cos x \\
&\implies e^y - ye^{-y}dy = 2\sin x dx \\
&\implies \int e^y - ye^{-y}dy = \int 2\sin x dx + C \\
&\implies \int e^y dy - \int ye^{-y}dy = \int 2\sin x dx + C
\end{aligned}$$

Using integration by parts for $\int ye^{-y}dy$, this implies

$$\int ye^{-y}dy = -ye^{-y} - e^{-y}$$

Therefore,

$$\begin{aligned}
&\int e^y dy - \int ye^{-y}dy = \int 2\sin x dx + C \\
\implies & e^y + ye^{-y} + e^{-y} = -2\cos x + C \\
\implies & e^y + ye^{-y} + e^{-y} + 2\cos x = C,\ C \in \mathbb{R}
\end{aligned}$$

Then $e^y + ye^{-y} + e^{-y} + 2\cos x = C$ is a general solution for constant C.
Let $y(0) = 0$, thus we obtain

$$1 + 0 + 1 + 2 = C = 4.$$

Hence $e^y + ye^{-y} + e^{-y} + 2\cos x = 4$ is a solution with IVP.

2.4 Exact Equations

Recall the differential:

Definition. The form

$$\Delta y \approx f(x+h) - f(x) = f'(x)dx = dy$$

is the differential of y.

Definition. For the function $z = f(x,\, y)$, the total differential is defined as

$$dz = df(x,\, y) = \frac{\partial f(x,\, y)}{\partial x}dx + \frac{\partial f(x,\, y)}{\partial y}dy$$

Remark. If $df(x,\, y) = \dfrac{\partial f(x,\, y)}{\partial x}dx + \dfrac{\partial f(x,\, y)}{\partial y}dy = 0$, then $f(x,\, y) = c$ is a solution.

Example. Solve $2xy\,dx + (x^2 - 1)\,dy = 0$.

Sol. Let

$$\begin{cases} 2xy & = \dfrac{\partial f(x,\, y)}{\partial x} \\ x^2 - 1 & = \dfrac{\partial f(x,\, y)}{\partial y} \end{cases}$$

Since $2xy = \dfrac{\partial f(x,\, y)}{\partial x}$, this implies

$$f(x,\, y) = \int 2xy\,dx = x^2y + g(y).$$

Thus

$$\begin{aligned} x^2 - 1 = \frac{\partial f(x,\, y)}{\partial y} &= \frac{\partial}{\partial y}\left(x^2y + g(y)\right) \\ \implies x^2 - 1 &= x^2 + g'(y) \end{aligned}$$

Therefore, we obtain $g'(y) = -1$, then $g(y) = -y$.
Hence $f(x,\, y) = x^2y - y = C$, $C \in \mathbb{R}$.

Example. Solve

$$M(x,\, y)\,dx + N(x,\, y)\,dy = 0.$$

Sol. Let

$$\begin{cases} M(x,\, y) & = \dfrac{\partial f(x,\, y)}{\partial x} \\ N(x,\, y) & = \dfrac{\partial f(x,\, y)}{\partial y} \end{cases}$$

Thus

$$\begin{cases} \dfrac{\partial}{\partial y}\left(\dfrac{\partial f(x,\,y)}{\partial x}\right) = \dfrac{\partial M(x,\,y)}{\partial y} \\ \dfrac{\partial}{\partial x}\left(\dfrac{\partial f(x,\,y)}{\partial y}\right) = \dfrac{\partial N(x,\,y)}{\partial x} \end{cases}$$

What if $\dfrac{\partial M(x,\,y)}{\partial y} = \dfrac{\partial N(x,\,y)}{\partial x}$?

Remark. An ODE $M(x,\,y)\,dx + N(x,\,y)\,dy = 0$ is **exact** if

$$\frac{\partial M(x,\,y)}{\partial y} = \frac{\partial N(x,\,y)}{\partial x}.$$

Example. Back to the example $2xy\,dx + (x^2 - 1)\,dy = 0$.
Let

$$M(x,\,y) = 2xy \text{ and } N(x,\,y) = x^2 - 1.$$

Thus

$$\begin{aligned} \frac{\partial M(x,\,y)}{\partial y} &= \frac{\partial (2xy)}{\partial y} = 2x \\ \frac{\partial N(x,\,y)}{\partial x} &= \frac{\partial\,(x^2 - 1)}{\partial y} = 2x \end{aligned}$$

Then $\dfrac{\partial M(x,\,y)}{\partial y} = \dfrac{\partial N(x,\,y)}{\partial x} = 2x$, the equation is exact.

Example. Solve $(e^{2y} - y\cos xy)\,dx + (2xe^{2y} - x\cos xy + 2y)\,dy = 0$.

Sol. Let $M(x,\,y) = e^{2y} - y\cos xy$ and $N(x,\,y) = 2xe^{2y} - x\cos xy + 2y$.
Then

$$\begin{aligned} \frac{\partial M(x,\,y)}{\partial y} &= \frac{\partial}{\partial y}e^{2y} - y\cos xy = 2e^{2y} - \cos xy + xy\sin xy \\ \frac{\partial N(x,\,y)}{\partial x} &= \frac{\partial}{\partial x}2xe^{2y} - x\cos xy + 2y = 2e^{2y} - \cos xy + xy\sin xy \end{aligned}$$

We obtain

$$\frac{\partial M(x,\,y)}{\partial y} = \frac{\partial N(x,\,y)}{\partial x} = 2e^{2y} - \cos xy + xy\sin xy,$$

the equation is exact. Let

$$\begin{cases} \dfrac{\partial f(x,\,y)}{\partial x} &= M(x,\,y) = e^{2y} - y\cos xy \\ \dfrac{\partial f(x,\,y)}{\partial y} &= N(x,\,y) = 2xe^{2y} - x\cos xy + 2y, \end{cases}$$

Since $\dfrac{\partial f(x,\,y)}{\partial x} = e^{2y} - y\cos xy$, thus

$$\begin{aligned} f(x,\,y) &= \int e^{2y} - y\cos xy\,dx + g(y) \\ &= xe^{2y} - \frac{y}{y}\sin xy + g(y) \\ &= xe^{2y} - \sin xy + g(y) \end{aligned}$$

Therefore,

$$\begin{aligned} \frac{\partial f(x,\,y)}{\partial y} &= 2xe^{2y} - x\cos xy + 2y \\ &= 2xe^{2y} - x\cos xy + g'(y). \end{aligned}$$

Then

$$2xe^{2y} - x\cos xy + 2y = 2xe^{2y} - x\cos xy + g'(y) \iff 2y = g'(y).$$

This implies

$$g(y) = y^2.$$

Thus we have

$$f(x,\,y) = xe^{2y} - \sin xy + y^2.$$

Hence

$$xe^{2y} - \sin xy + y^2 = C$$

is a solution for constant C.

Example. (Exercise) Solve $\dfrac{dy}{dx} = \dfrac{xy^2 - \cos x\sin x}{y\,(1-x^2)}$ with $y(0) = 2$.

Sol.

$$\begin{aligned} \frac{dy}{dx} = \frac{xy^2 - \cos x\sin x}{y(1-x^2)} &\implies y(1-x^2)dy = \left(xy^2 - \cos x\sin x\right)dx \\ &\implies \left(xy^2 - \cos x\sin x\right)dx - y\left(1-x^2\right)dy = 0 \\ &\implies \left(xy^2 - \cos x\sin x\right)dx + y\left(x^2-1\right)dy = 0 \end{aligned}$$

Let $M(x,\,y) = xy^2 - \cos x\sin x$ and $N(x,\,y) = y\,(x^2-1)$, then compute

$$\frac{\partial M}{\partial y} = 2xy \text{ and } \frac{\partial N}{\partial x} = 2xy.$$

Since $\dfrac{\partial M}{\partial y} = \dfrac{\partial N}{\partial x} = 2xy$, the equation is exact. Thus

$$\begin{cases} \dfrac{\partial f(x,\,y)}{\partial x} = M(x,\,y) = xy^2 - \cos x\sin x \\ \dfrac{\partial f(x,\,y)}{\partial y} = N(x,\,y) = y\,(x^2-1) \end{cases}$$

So

$$\begin{aligned} f(x,\, y) &= \int y\left(x^2-1\right) dy \\ &= \frac{1}{2}y^2\left(x^2-1\right) + h(x) \\ \implies \frac{\partial f(x,\, y)}{\partial x} &= \frac{\partial}{\partial x}\left(\frac{1}{2}y^2\left(x^2-1\right) + h(x)\right) \\ &= xy^2 + h'(x) \end{aligned}$$

Since $\dfrac{\partial f(x,\, y)}{\partial x} = xy^2 + h'(x) = xy^2 - \cos x \sin x$, we obtain $h'(x) = -\cos x \sin x$. Therefore,

$$\begin{aligned} h'(x) &= -\cos x \sin x \\ \implies h(x) &= \frac{\cos^2 x}{2} \text{ (By } u\text{-substitution)} \end{aligned}$$

Then $f(x,\, y) = \dfrac{1}{2}y^2\left(x^2-1\right) + \dfrac{\cos^2 x}{2}$, that is $y^2\left(1-x^2\right) - \cos^2 x = C$ is a solution. Let $y(0) = 2$, we have

$$4(1-0) - 1 = 3 = C.$$

Hence

$$y^2\left(1-x^2\right) - \cos^2 x = 3$$

is a solution.

2.5 Solutions by Substitutions

Example. Is $2xyy' = y^2 - x^2$ separable? The answer is **NO**!

Example. From last example, we have

$$\begin{aligned} 2xy\frac{dy}{dx} = y^2 - x^2 &\implies \frac{dy}{dx} = \frac{y^2}{2xy} - \frac{x^2}{2xy} \\ &\implies \frac{dy}{dx} = \frac{y}{2x} - \frac{x}{2y} \end{aligned}$$

It can't solve by separable method, but if we let $u = \dfrac{y}{x}$, we obtain

$$\frac{dy}{dx} = \frac{y}{2x} - \frac{x}{2y} = \frac{1}{2}\left(u - \frac{1}{u}\right)$$

Since $u = \dfrac{y}{x}$, that is $y = ux$, then by chain rule and product rule,

$$\frac{dy}{dx} = \frac{du}{dx}x + u.$$

Therefore,

$$\begin{aligned} \frac{dy}{dx} = \frac{1}{2}\left(u - \frac{1}{u}\right) &\implies \frac{du}{dx}x + u = \frac{1}{2}\left(u - \frac{1}{u}\right) \\ &\implies x\frac{du}{dx} = \frac{-(1+u^2)}{2u} \\ &\implies \frac{2u}{(1+u^2)}du = \frac{1}{x}dx \\ &\implies \int \frac{2u}{(1+u^2)}du = \int \frac{-1}{x}dx + C \\ &\implies \ln\left(1+u^2\right) = -\ln|x| + C \\ &\implies 1 + u^2 = \left|\frac{1}{x}\right| + e^C. \end{aligned}$$

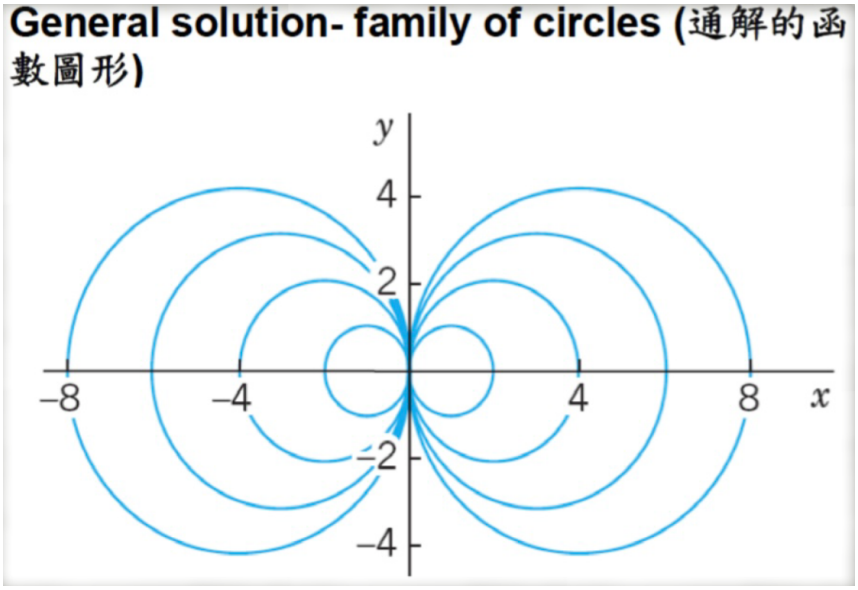
Then take $u = \dfrac{y}{x}$ back to the equation

$$1 + \left(\frac{y}{x}\right)^2 = \frac{1}{x} + e^C \implies x^2 + y^2 = xe^C,\ C \in \mathbb{R},$$

also we can let $a = e^C$ then we obtain

$$x^2 + y^2 = xa \implies \left(x - \frac{a}{2}\right)^2 + y^2 = \frac{a^2}{4}.$$

Hence $\left(x - \dfrac{a}{2}\right)^2 + y^2 = \dfrac{a^2}{4}$ is a general solution.



Recall the chain rule for partial derivative on $u = f(x, y) \implies y = h(x, u)$:

$$dy = \frac{\partial y}{\partial x}dx + \frac{\partial y}{\partial u}du$$

Example. Solve

$$(x^2 + y^2)dx + (x^2 - xy)dy = 0.$$

Sol. Let $u = \dfrac{y}{x}$, then $y = ux$. Compute dy, we have

$$\begin{aligned} dy &= \frac{\partial y}{\partial x}dx + \frac{\partial y}{\partial u}du \\ &= udx + xdu \end{aligned}$$

Thus

$$\begin{aligned} & (x^2 + y^2)dx + (x^2 - xy)dy = 0 \\ \implies & (x^2 + u^2x^2)dx + (x^2 - ux^2)(udx + xdu) = 0 \\ \implies & x^2(1 + u^2)dx + x^2(u - u^2)dx + x^3(1 - u)du = 0 \\ \implies & x^2(1 + u)dx + x^3(1 - u)du = 0 \\ \implies & \frac{1}{x}dx + \frac{1 - u}{1 + u}du = 0. \end{aligned}$$

Then

$$\begin{aligned}
\frac{1-u}{1+u}du = -\frac{1}{x}dx &\implies \int \frac{1-u}{1+u}du = \int -\frac{1}{x}dx + C \\
&\implies \int \frac{1}{1+u}du - \int \frac{u}{1+u}du = \ln\left|\frac{1}{x}\right| + C \\
&\implies \ln|1+u| - (u - \ln|1+u|) = \ln\left|\frac{1}{x}\right| + C \\
&\implies 2\ln|1+u| - u = \ln\left|\frac{1}{x}\right| + C \\
&\implies \ln(1+u)^2 - u - \ln\left|\frac{1}{x}\right| = C,\ C \in \mathbb{R}.
\end{aligned}$$

Hence $\ln(1+u)^2 - u - \ln\left|\frac{1}{x}\right| = C$ is a general solution for constant C.